Hannah's Lab Results — June 25, 2026

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Patient: Hannah Munguia-Flores | **DOB:** 8/18/1995 | **MRN:** 10161495501
PCP / Ordering Provider: Kristen Elaine Medley, MD, MPA
Resulting Lab: Brigham and Women’s Faulkner Hospital, 1153 Centre Street, Boston, MA 02130 (CLIA #22D0663738)
Lab Director: Agoston Agoston, MD, PhD
Collection Date (all panels): Jun 25, 2026 at 10:37 AM
Status: All results Final

⚠ FLAGGED / ABNORMAL VALUES AT A GLANCE

Test	Result	Units	Reference Range	Flag
Ferritin	**191**	ug/L	30 – 150	🔴 **HIGH**
Triglycerides	**355**	mg/dL	≤ 150	🔴 **HIGH**

> All other values were within normal reference ranges.

Panel 1 — LFTs (Hepatic Panel)

Result Date: Jun 25, 2026 12:41 PM | **Source file:** MGB 1 (also MGB 6)

Test	Result	Units	Reference Range	Flag
AST	20	U/L	10 – 50	Normal
ALT	10	U/L	10 – 50	Normal
Alkaline Phosphatase	40	U/L	40 – 130	Normal
Bilirubin, Total	0.2	mg/dL	0.0 – 1.2	Normal
Bilirubin, Direct	0.1	mg/dL	0.0 – 0.3	Normal
Total Protein	7.6	g/dL	6.4 – 8.3	Normal
Albumin	3.9	g/dL	3.5 – 5.2	Normal
Globulin	3.7	g/dL	1.9 – 4.1	Normal

Panel 2 — Ferritin

Result Date: Jun 25, 2026 12:41 PM | **Source file:** MGB 2

Test	Result	Units	Reference Range	Flag
Ferritin	**191**	ug/L	30 – 150	**🔴 H (HIGH)**

> *Note: Lab updated the lower limit of the reference range to 30 ug/L for all adults to reflect a physiologically sufficient level (was previously lower).*

Panel 3 — Iron and Iron Binding Capacity

Result Date: Jun 25, 2026 12:41 PM | **Source file:** MGB 2

Test	Result	Units	Reference Range	Flag
Iron	92	ug/dL	28 – 170	Normal
Total Iron-Binding Capacity (TIBC)	347	ug/dL	220 – 460	Normal
Transferrin Saturation	27	%	14 – 50	Normal

Panel 4 — 25-OH Vitamin D

Result Date: Jun 25, 2026 12:44 PM | **Source file:** MGB 2

Test	Result	Units	Reference Range	Flag
25-OH Vitamin D, Total	49	ng/mL	20 – 50	Normal

> *Interpretive scale:*
> - Severe deficiency: < 10 ng/mL
> - Mild-moderate deficiency: 10–19 ng/mL
> - **Optimum: 20–50 ng/mL** ← Hannah's result falls here
> - Increased risk of hypercalciuria: 51–80 ng/mL
> - Possible toxicity: > 80 ng/mL

Panel 5 — TSH with Reflex

Result Date: Jun 25, 2026 12:35 PM | **Source files:** MGB 2, MGB 5

Test	Result	Units	Reference Range	Flag
TSH	3.32	uIU/mL	0.40 – 5.00	Normal

> *No reflex testing triggered (TSH within normal range).*

Panel 6 — Basic Metabolic Panel (BMP)

Result Date: Jun 25, 2026 12:41 PM | **Source files:** MGB 3, MGB 6

Test	Result	Units	Reference Range	Flag
Sodium	136	mmol/L	136 – 145	Normal (low-normal

				boundary)
Potassium	4.1	mmol/L	3.4 – 5.1	Normal
Chloride	101	mmol/L	98 – 107	Normal
CO2	27	mmol/L	20 – 31	Normal
BUN	10	mg/dL	6 – 23	Normal
Creatinine	0.73	mg/dL	0.50 – 1.00	Normal
Glucose	84	mg/dL	70 – 99	Normal
Calcium	9.4	mg/dL	8.5 – 10.5	Normal
eGFR	113	mL/min/1.73m^2	> 59	Normal
Anion Gap	8	mmol/L	3 – 17	Normal

> *Note: eGFR calculated using CKD-EPI refit equation. Sodium at 136 is exactly at the lower boundary of normal — not flagged, but worth noting.*

Panel 7 — CBC (Complete Blood Count)

Result Date: Jun 25, 2026 11:56 AM | **Source files:** MGB 3, MGB 7, MGB 9

Test	Result	Units	Reference Range	Flag
WBC	8.43	K/uL	4.00 – 11.00	Normal
RBC	4.22	M/uL	4.00 – 5.20	Normal
Hemoglobin	12.8	g/dL	12.0 – 16.0	Normal
Hematocrit	38	%	36.0 – 46.0	Normal
MCV	90	fL	80.0 – 100.0	Normal
MCH	30.3	pg	27.0 – 31.0	Normal
MCHC	33.7	g/dL	32.0 – 36.0	Normal
Platelets (PLT)	324	K/uL	150 – 450	Normal
MPV	10.5	fL	8.4 – 12.0	Normal
RDW-CV	12	%	11.5 – 14.5	Normal
Absolute NRBC	0	K cells/uL	≤ 0.00	Normal
NRBC	0	/100 WBCs	≤ 0.0	Normal

Panel 8 — Lipid Panel

Result Date: Jun 25, 2026 12:41 PM | **Source files:** MGB 4, MGB 10

Test	Result	Units	Reference Range	Flag
Cholesterol, Total	180	mg/dL	< 200	Normal
HDL	59	mg/dL	≥ 40	Normal
LDL (calculated)	66	mg/dL	< 130	Normal
Non-HDL Cholesterol	121	mg/dL	Goal: LDL goal + 30	Not flagged
Cardiac Risk Ratio	3.1	ratio	0.0 – 5.0	Normal
Triglycerides	**355**	mg/dL	≤ 150	🔴 **H (HIGH)**

> *LDL calculated using the Sampson-NIH equation (JAMA Cardiol. 2020 May 1;5(5):540-548).*
> *Triglycerides of 355 mg/dL is markedly elevated — more than 2× the upper limit of normal. This is clinically significant.*

Panel 9 — Hemoglobin A1c

Result Date: Jun 25, 2026 12:08 PM | **Source files:** MGB 4, MGB 11

Test	Result	Units	Reference Range	Flag
Hemoglobin A1c (HbA1c)	5.2	%	4.3 – 5.6	Normal
Calculated Mean Blood Glucose (EAG)	103	mg/dL	No established range	—

> *Interpretive thresholds for HbA1c:*
> - Normal: 4.3–5.6% (EAG ≤ 114 mg/dL at 5.6%)
> - Prediabetes: 5.7–6.4%
> - Diabetes: ≥ 6.5% (EAG ≥ 140 mg/dL)

File-to-Panel Mapping

File	Panel(s)
MGB 1 (combined-name file)	LFTs (Hepatic Panel)
MGB 2	Ferritin, Iron/TIBC, Vitamin D, TSH
MGB 3	Basic Metabolic Panel, CBC
MGB 4	Lipid Panel, HbA1c
MGB 5	TSH (standalone)
MGB 6	LFTs, Basic Metabolic Panel (duplicate)
MGB 7	CBC (duplicate)
MGB 9	CBC (duplicate)
MGB 10	Lipid Panel (duplicate)
MGB 11	HbA1c (duplicate)

> *Note: MGB 8 was not included in the provided file set. Several files appear to be duplicates of earlier reports from the same draw.*

Complete Test Count

Category	Tests	Status
LFTs (Hepatic Panel)	8	✅ All normal
Iron studies (Ferritin, Iron, TIBC, Transferrin Sat)	4	⚠ Ferritin HIGH
Vitamin D	1	✅ Normal
TSH	1	✅ Normal
		✅ All normal (Na

Basic Metabolic Panel	10	low-normal)
CBC	12	✅ All normal
Lipid Panel	6	⚠ Triglycerides HIGH
HbA1c	2	✅ Normal
Total unique tests	**44**	**2 abnormal flags**

Clinical Notes

🔴 Ferritin 191 ug/L (HIGH — ref: 30–150 ug/L)

Elevated ferritin can reflect: - Acute phase reaction / inflammation (very common in Long COVID) - Iron overload / hemochromatosis (less likely given normal iron/TIBC/transferrin saturation) - Liver disease (LFTs normal, so unlikely) - Other inflammatory or metabolic conditions

Given that iron, TIBC, and transferrin saturation are **all normal**, this pattern (elevated ferritin, normal iron indices) is most consistent with **inflammatory ferritin elevation**, which is commonly seen in Long COVID and other inflammatory conditions. Not indicative of iron overload.

🔴 Triglycerides 355 mg/dL (HIGH — ref: ≤150 mg/dL)

Markedly elevated. Note that other lipid values are favorable (LDL 66, HDL 59, total cholesterol 180). Elevated triglycerides in isolation can be associated with: - Dietary factors (fasting status at draw time may matter) - Metabolic syndrome features (though glucose and HbA1c are normal) - Medication effects - Genetic hypertriglyceridemia - Hypothyroidism (TSH is normal)

Clinically worth following up. At > 500 mg/dL, pancreatitis risk rises; at 355, clinical management discussion is appropriate.

✅ Notably Reassuring

- **HbA1c 5.2%** — no prediabetes signal despite elevated triglycerides
- **Creatinine 0.73, eGFR 113** — kidney function excellent
- **LFTs all normal** — no hepatic inflammation
- **CBC all normal** — no anemia, thrombocytopenia, or leukocytosis
- **TSH 3.32** — thyroid function normal
- **Vitamin D 49 ng/mL** — optimal range (likely supplementing successfully)
- **HDL 59** — good protective cholesterol

Extracted by Cleo from 10 MGB PDF lab reports on 2026-06-25. Source: Brigham and Women's Faulkner Hospital lab system, single blood draw Jun 25, 2026 at 10:37 AM.